

Marcos Skowronek Santos

Department of Physics, Brown University, Providence, RI 02912, USA | marcos_skowronek_santos@brown.edu
(401) 255-3127

Education

Ph.D. , Physics: Brown University	September 2023 – Present
M.Sc. , Physics: University of Copenhagen	September 2021 – June 2023
B.A. , Physics: Universidad Complutense de Madrid (Extraordinary Bachelor's Award)	September 2017 – June 2021

Professional appointments

Research and Teaching Assistant, Brown University	September 2023 – Present
Student Assistant, Niels Bohr International Academy	August 2022 – June 2023
Research Assistant, Universidad Complutense de Madrid	September 2020 – June 2021
	June 2005 – Aug 2007

Academic Honors

1. 2026-2027 Physics Dissertation Fellowship, Brown University, 2026.
2. Undergraduate Extraordinary Award, Universidad Complutense de Madrid, 2021.
3. Award for Excellence in Academic Performance (Autonomous Community of Madrid), 2021.
4. Award for Excellence in Academic Performance (Autonomous Community of Madrid), 2020.
5. Award for Excellence in Academic Performance (Autonomous Community of Madrid), 2019.
6. Award for Excellence in Academic Performance (Autonomous Community of Madrid), 2018.
7. Award for the 100 best scores in the university access exam (Universidad Complutense de Madrid), 2017.
8. Award for the 100 best scores in the university admission process (Universidad Complutense de Madrid), 2017.

Research Grants

1. SU grant, The Danish Ministry of Higher Education and Science, 2022.
2. Stefan Rozental og Hanna Kobylinski Rozentals Fond, Røvsing & Gammeljord, 2022.
3. Collaboration Grant, Department of Theoretical Physics, Universidad Complutense de Madrid (Spanish Ministry of Education), 2020.

Publications

1. C. R. T. Jones, S. Paranjape and M. Skowronek, “Bound States in Perturbative Quantum Gravity with Hydrogen-like Degeneracy,” [arXiv:2608.29402 [hep-th]].
2. S. Paranjape, M. Skowronek, M. Spradlin, A. Volovich and H. C. Weng, “Landau Analysis of One-Cycle Negative Geometries,” [arXiv:2604.22683 [hep-th]].
3. S. Paranjape, M. Skowronek, M. Spradlin, A. Volovich and H. C. Weng, “Cluster Bootstrap for Cosmological Correlators,” JHEP **2026**, 121 (2026) doi:10.1007/JHEP09(2026)121 [arXiv:2603.08670 [hep-th]].
4. C. Figueiredo and M. Skowronek, “Cuts and Contours,” JHEP **12**, 024 (2025) doi:10.1007/JHEP12(2025)024 [arXiv:2506.05456 [hep-th]].
5. S. Paranjape, M. Skowronek, M. Spradlin and A. Volovich, “Large deformations of $\text{Tr}(\Phi^3)$ and the world at infinity,” JHEP **01**, 076 (2026) doi:10.1007/JHEP01(2026)076 [arXiv:2504.11253 [hep-th]].

6. S. De, A. Pokraka, M. Skowronek, M. Spradlin and A. Volovich, “Surfaceology for colored Yukawa theory,” JHEP **09**, 160 (2024) doi:10.1007/JHEP09(2024)160 [arXiv:2406.04411 [hep-th]].
7. N. E. J. Bjerrum-Bohr, G. Chen, Y. Miao and M. Skowronek, “Color-kinematic numerators for fermion Compton amplitudes,” JHEP **07**, 242 (2024) doi:10.1007/JHEP07(2024)242 [arXiv:2404.15265 [hep-th]].
8. N. E. J. Bjerrum-Bohr, G. Chen and M. Skowronek, “Covariant Compton Amplitudes in Gravity with Classical Spin,” Phys. Rev. Lett. **132**, no.19, 191603 (2024) doi:10.1103/PhysRevLett.132.191603 [arXiv:2309.11249 [hep-th]].
9. N. E. J. Bjerrum-Bohr, G. Chen and M. Skowronek, “Classical spin gravitational Compton scattering,” JHEP **06**, 170 (2023) doi:10.1007/JHEP06(2023)170 [arXiv:2302.00498 [hep-th]].
10. J. A. R. Cembranos and M. Skowronek, “Functional Quantum Field Theory in Phase Space,” [arXiv:2108.06268 [hep-th]].

To appear

1. N. Arkani-Hamed, C. Figueiredo, M. Skowronek, “The AdS/Fishnet Correspondence”.
2. N. Arkani-Hamed, C. Figueiredo, S. Paranjape, M. Skowronek, “The World at Infinity for Stringy Integrals”.

Invited talks

1. *Prospects in Theoretical Physics 2026*, Institute for Advanced Study, July 2026.
2. *Amplitudes 2026 Conference*, Queen Mary University of London, June 2026.
3. *Symbology@15*, Max Planck Institute for Physics, December 2025.
4. HET seminar, Max Planck Institute for Physics, December 2025.
5. *Michigan Interdisciplinary Meeting on Amplitudes: Bridges between Physics & Mathematics*, University of Michigan, October 2025.
6. *TASI 2025* (student talk), University of Colorado Boulder, June 2025.
7. *Amplitudes 2024 Conference and Summer School*, Institute for Advanced Study, June 2024.

Attended schools and conferences

1. *Prospects in Theoretical Physics 2026*, Institute for Advanced Study, July 2026.
2. *Amplitudes 2026 Summer School*, University of Southampton, July 2026.
3. *Amplitudes 2026 Conference*, Queen Mary University of London, June 2026.
4. *Lancefest Workshop*, University of Edinburgh, June 2026.
5. *Max Planck-IAS-NTU Center Kick-off Conference*, Institute for Advanced Study, March 2026.
6. *Symbology@15*, Max Planck Institute for Physics, December 2025.
7. *Michigan Interdisciplinary Meeting on Amplitudes: Bridges between Physics & Mathematics*, University of Michigan, October 2025.
8. *TASI 2025*, University of Colorado Boulder, June 2025.
9. *Combinatorics for Fundamental Physics Workshop*, Institute for Advanced Study, November 2024.
10. *Amplitudes 2024 Conference and Summer School*, Institute for Advanced Study, June 2024.
11. *Amplituhedra, Cluster Algebras and Positive Geometry Conference*, Center of Mathematical Sciences and Applications, Harvard University, May 2024.

12. *Program on Mathematical Aspects of Scattering Amplitudes*, Center of Mathematical Sciences and Applications, Harvard University, April - May 2024.
13. *Kickoff Workshop for the Simons Collaboration in Celestial Holography*, Harvard University, October 2023.
14. *Jumpstarting Elliptic Bootstrap Methods for Scattering Amplitudes*, Niels Bohr Institute, September 2022.
15. *Bohr-100: Current Themes in Theoretical Physics*, Niels Bohr Institute, August 2022.

Research travel

1. Institute for Advanced Study, March 2026, invited by Prof. Nima Arkani-Hamed.
2. Max Planck Institute for Physics, December 2025, invited by Prof. Johannes Henn.
3. Institute for Advanced Study, July 2025, invited by Prof. Nima Arkani-Hamed.
4. Institute for Advanced Study, April 2025, invited by Prof. Nima Arkani-Hamed.
5. Institute for Advanced Study, March 2025, invited by Prof. Nima Arkani-Hamed.
6. Institute for Advanced Study, September 2024, invited by Prof. Nima Arkani-Hamed.

Teaching

Brown University:

- F2023: PHYS0030, Basic Physics A.
- S2024: PHYS0030, Basic Physics A.
- F2024: PHYS2030, Classical Theoretical Physics I.
- S2025: PHYS2040, Classical Theoretical Physics II.